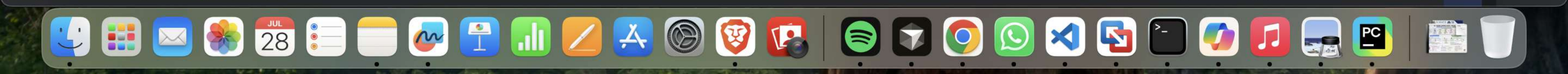


```
1 grammar = {'S': [['NP', 'VP']], 'NP': [['Det', 'N']], 'VP': [['V', 'NP']], 'Det': [['the'], ['a']], 'N': [['cat'], ['dog']], 'V': [['sees'], ['likes']]}
2 sentence = input("Enter sentence: ").lower().split()
3 def parse(symbol, words, pos): 2 usages
4     if symbol not in grammar:
5         if pos < len(words) and symbol == words[pos]:
6             return (symbol, []), pos + 1
7         return None, pos
8     for rule in grammar[symbol]:
9         children = []
10        current = pos
11        success = True
12        for item in rule:
13            tree, current = parse(item, words, current)
14            if tree is None:
15                success = False
16                break
17            children.append(tree)
18        if success:
19            return (symbol, children), current
20    return None, pos
21 def print_tree(tree, level=0): 2 usages
22     symbol, children = tree
23     print(" " * level + symbol)
24     for child in children:
25         print_tree(child, level + 1)
26 tree, position = parse(symbol: 'S', sentence, pos: 0)
```

Run main

Process finished with exit code 1



```
27 if tree and position == len(sentence):
28     print("\nParse Tree:")
29     print_tree(tree)
30 else:
31     print("Sentence cannot be parsed")
```

Run main

/Users/moshika/PyCharmMiscProject/.venv/bin/python /Users/moshika/PyCharmMiscProject/Exp 13.py
Enter sentence: *the cat sees a dog*

Parse Tree:
S
 NP
 Det
 the
 N
 cat
 VP
 V
 sees
 NP
 Det
 a
 N
 dog

Process finished with exit code 0

